COMPARISON OF R290 AS AN ALTERNATIVE REFRIGERANT FOR REPLACING R32 IN A DOMESTIC SPLIT TYPE AIR CONDITIONER

PROJECT REPORT

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DECLARATION

We hereby declare that the work presented in this report titled "Comparison of R290 as an Alternative Refrigerant for Replacing R32 in a Domestic Split Type Air Conditioner" is an original piece of work carried out by group RvsR as part of the project requirements for the degree program at the Faculty of Engineering, University of Peradeniya.

All sources of information, ideas, and data that have been referred to in this report are properly acknowledged and referenced. The project has not been submitted for any other degree or diploma, and the results of this work have not been previously published elsewhere.

We confirm that all the relevant ethical considerations, including the safe handling of refrigerants, have been adhered to during the course of this project. The information provided in this report is accurate and true to the best of our knowledge.

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APPROVAL/ CERTIFICATION

ABSTRACT

This study aims to compare R32 and R290 as refrigerants for domestic split air conditioning systems. The research work will be focused on the environmental impact, energy efficiency, safety considerations, and cost of replacement of R32 with R290. R290 is a natural refrigerant with a very low GWP and zero ODP, hence offering a significant environmental benefit over R32. However, since it is a flammable refrigerant, system retrofits are performed with great precaution, including various safety features like flameproof elements and leak detectors. Energy efficiency analysis has proved that, due to thermodynamically favorable properties of R290, it offers the same, if not higher, level of cooling at lower consumption of energy. Although the R290 systems could be slightly more expensive initially, the after operational savings would offset most of the environmental costs involved. In practice, this replacement would involve measuring the pressures and temperatures of the condenser and evaporator in a controlled climate chamber; the cooling capacity would be calculated based on temperature differences over time. Results show that R290 is a viable, sustainable, and energy-efficient alternative to R32 for greener and more cost-effective cooling technologies.

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LIST OF SYMBOLS AND ABBREVIATIONS

ODP	Ozone Depletion Potential
GWP	Global Warming Potential
AC	Air Conditioning
COP	Coefficient of Performance
HFC	Hydroflurocarbon
CFC	Chloroflurocarbon
ISO	International Organization for Standardization
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers

NOTATION AND CLASSIFICATION

R-32	Difluoromethane (CH_2F_2)
R-290	Propane (C_3H_8)
R-22	Chlorodifluoromethane ($CHClF_2$)
R-410a	Puron (Blend of Difluromethane & Pentafluroethane)
R-407c	mixture of hydrofluorocarbons
R-134a	Tetrafluoroethane
P	Pressure
S	Entropy
h	Enthalpy

CHAPTER 1

INTRODUCTION

1.1 Motivation

The urgent need to mitigate climate change has spurred significant advancements in sustainable technologies across various industries. One critical area of focus is the reduction of greenhouse gas emissions, particularly those associated with refrigerants used in air conditioning systems. Halogenated refrigerants, such as R32, have been widely utilized but are now recognized for their detrimental impact on the environment, contributing to ozone depletion and global warming.

In response to this pressing environmental challenge, the research community has turned its attention to alternative refrigerants that offer superior eco-friendliness without compromising performance and efficiency. Among these alternatives, R290, or propane, has emerged as a promising candidate due to its favorable environmental profile and thermodynamic properties.

The motivation for this project stems from the imperative to transition away from halogenated refrigerants like R32 towards more sustainable options such as R290 by, conducting a comprehensive comparison between R32 and R290.

Performance Analysis:

Investigate the thermodynamic properties and performance characteristics of R32 and R290 in air conditioning systems. This includes assessing factors such as cooling capacity, energy efficiency, and operational reliability under varying operating conditions.

Economic Viability:

Examine the economic feasibility of transitioning from R32 to R290, taking into account factors such as initial investment costs, lifecycle costs, and regulatory compliance. A cost-benefit analysis will help stakeholders make informed decisions regarding the adoption of R290 as a replacement refrigerant.

Safety Considerations:

Explore the safety aspects associated with the use of R290 as a refrigerant, including flammability risks and mitigation strategies. Understanding the potential hazards and implementing appropriate safety measures are essential for ensuring the safe deployment of R290-based air conditioning systems.

Environmental Impact Assessment:

Evaluate the environmental footprint of R32 and R290, considering factors such as ozone depletion potential (ODP) and global warming potential (GWP). This assessment will provide insights into the overall sustainability of each refrigerant and their contribution to climate change.

By addressing these factors, our project aims to provide valuable insights into the suitability of R290 as a viable alternative to R32 in air conditioning systems. Ultimately, the findings of this study will contribute to the ongoing efforts to promote sustainability and reduce the environmental impact of refrigeration technologies, facilitating the transition towards a greener and more sustainable future.

1.2 Project Overview

This project involves a detailed comparative analysis between R32 and R290 refrigerants in the context of domestic split-type air conditioning systems. The comparison will encompass various parameters including energy efficiency, safety considerations, cost comparison and environmental impact. Experimental investigations will be conducted to measure the performance characteristics of both refrigerants under controlled conditions, simulating real-world operating environments of air conditioning units. Additionally, theoretical modeling and simulations will complement the experimental findings to provide a comprehensive understanding of the comparative advantages and limitations of each refrigerant. The project will culminate in a conclusive assessment highlighting the suitability of R290 as an alternative refrigerant for replacing R32 in domestic air conditioning applications.

The background of this project stems from the need to address the environmental concerns associated with conventional refrigerants used in air conditioning systems,

particularly R32, which has a relatively high GWP. As regulatory measures tighten and environmental awareness grows, there is a critical imperative for the AC industry to transition towards low-GWP alternatives. R290, a hydrocarbon refrigerant with significantly lower GWP and excellent thermodynamic properties, emerges as a promising candidate for sustainable cooling solutions. However, its adoption necessitates a thorough understanding of its performance characteristics and safety considerations compared to R32. Hence, this project seeks to bridge this gap by providing an in-depth comparative analysis, laying the groundwork for informed decision-making in the transition towards more environmentally friendly refrigerants in domestic air conditioning systems.

1.3 Methodology

Literature Review:

- Research existing studies on R290 and R32, focusing on their thermodynamic properties, physical properties, efficiency, safety considerations and environmental impact.
- Identify industry standards and regulations related to refrigerants.

Experimental Setup:

- Unit Selection: Use a split-type AC unit specifically designed for R32. Ensure it’s compatible with R290 through necessary modifications or safeguards.
- Instrumentation: Equip the units with energy meters, temperature sensors, and pressure gauges for accurate data collection.

Performance Testing:

- Energy Efficiency:
    - Measure the coefficient of performance (COP) for both refrigerants under similar operating conditions.
    - Test at various ambient temperatures.
    - Measure power consumption, cooling capacity, and energy output.

- Cost Analysis:
    - Initial Cost: Compare the cost of R290 vs. R32 and any additional modifications needed for R290.
    - Operating Cost: Analyze energy costs based on efficiency data and local energy prices.
    - Maintenance Cost: Evaluate expected maintenance needs and associated costs for both systems.

Safety Evaluation:

- Assess the flammability and toxicity of R290 compared to R32.
- Review safety measures needed for handling, installation, and operation of both refrigerants.

Environmental Impact:

- Compare the global warming potential (GWP) and ozone depletion potential (ODP) of R290 and R32.

Data Analysis:

- Use statistical methods to analyze the collected data.
- Create graphs and tables to visually represent findings for easier comparison.

Reporting:

- Summarize findings in a clear, structured report.
- Include recommendations based on your comparisons and analyses.

Discussion:

- Discuss the implications of your findings on industry practices and regulations.
- Address limitations of the study and suggestions for future research.

Expected Outcomes

- ✓ Energy Efficiency: Clear metrics comparing COP and energy consumption.
- ✓ Cost-Benefit Analysis: A comprehensive comparison of both initial and operational costs.
- ✓ Safety Guidelines: Insights into safe handling practices for both refrigerants.

- ✓ Environmental Assessment: An evaluation of the ecological impact of both refrigerants.

1.4 Features of Project

This project deals with the assessment of R290 as a greener alternative to R32 in domestic split-type air conditioners based on energy efficiency, environmental effects, safety issues, and economic viability. The project intends to present a comparison of the performance of the two refrigerants through experimental analysis, considering the challenges involved in using R290 due to its flammability, while highlighting its environmentally friendly advantages. The result is expected to provide an indication of how R290 can be adopted in the refrigeration industry for sustainable engineering practices.

Research Methodology:

- Conducting experimental tests on air conditioners charged with R32 and R290.
- Measuring parameters such as pressure, temperature, energy consumption, and cooling capacity.
- Analyzing data to determine performance characteristics under various operating conditions.

Technical Components:

- Use of thermodynamic principles to compare refrigerants.
- Application of thermodynamic properties for evaluating cooling performance.
- Implementation of safety protocols when working with flammable refrigerants like R290.

Relevance to Industry:

- Direct application to domestic air conditioning systems.
- Contribution to the development of sustainable cooling technologies.

Academic Value:

- Integrates mechanical engineering principles in thermodynamics, and heat transfer.
- Enhances understanding of sustainable refrigerant alternatives for refrigeration applications.
- Opportunity to develop hands-on experimental and data analysis skills.

Challenges:

- Handling R290's flammability while ensuring safe and accurate testing.
- Balancing performance metrics with environmental and economic considerations.
- Overcoming potential resistance to adoption due to safety concerns.

Potential Impact:

- Providing recommendations for adopting R290 in the refrigeration industry.
- Demonstrating the feasibility of sustainable refrigerants in practical applications.
- Contributing to global efforts to reduce GWP in the cooling sector.

1.5 Project Plan

Proposed project timeline:

Week task	1	2	3	4	5	6	7	8	9	10	11	12	13	14
selection														
proposal submission														
proposal presentation														
literature survey														
data collection														
calculation and analysis														
final presentation and report submission														

Figure 01: Proposed project timeline

The timeline established for data collection indicated that an experimental analysis was to be conducted to compare the performance of the two refrigerants. However, the analysis could not be carried out due to the inability to procure the necessary instruments.

However, the field visit to AB manufacturing was done on 08/10/2024 and the existing air conditioning unit was studied. After the visit we identified that these two refrigerants work under different operating conditions and therefore, we cannot evaluate the COP values by considering the same evaporator and condenser temperatures.

Even though the experimental analysis could not be performed the theoretical analysis for the scenario was done under analyzing phase.

CHAPTER 2

LITERATURE SURVEY

2.1 What is R32?

R32 is a colorless gas that belongs to the HFC family of refrigerants. It is also known as difluoromethane. R32 is used as a refrigerant in air conditioning systems. It is a new generation of refrigerant that will efficiently circulate between indoor and outdoor units, while actively reducing electricity consumption up to as much as 10% compared to air conditioners using existing R-22 refrigerant. It is also far more energy efficient that other refrigerants on the market and has a global warming potential (GWP) that is two-thirds lower than comparable products. As an illustration the GWP of R32 is 675 compared to R410A (GWP 2088); R407C (GWP 1774); R134A (GWP 1430).

2.1.1 Features of R32

R32 is a refrigerant gas used as a substitute for Freon in air conditioners. This gas has a lower GWP than Freon and does not deplete the ozone layer, making it more environmentally friendly. Additionally, R32 does not deplete the ozone layer. R32 is a single blend HFC refrigerant on a countdown to phase out. Hydrocarbons are generally known to be eco-friendlier than HFC and CFC's, as they have a lower GWP and are more easily recyclable to be used again.

- Low Global Warming Potential (GWP): R-32 has a relatively low GWP of 675 over a 100-year period, contributing to reduced environmental impact compared to many other refrigerants.
- Single Component: R-32 is a single-component refrigerant composed of Difluoromethane, simplifying its handling and management.
- Energy Efficiency: It offers high energy efficiency, helping to reduce energy consumption and operating costs in air conditioning systems.
- Mildly Flammable: Classified as mildly flammable (A2L) according to ASHRAE standards, R-32 poses a lower flammability risk compared to other flammable refrigerants.

- Widely Used: R-32 is commonly used in residential and commercial air conditioning systems, particularly in split-type air conditioners.
- Regulatory Acceptance: It is recognized and accepted by regulatory authorities for use in air conditioning and refrigeration applications in many regions globally.

Table 01: Features of R32

Classification	Hydrofluorocarbon refrigerant
Chemistry	CH2F2
Ozone depletion potential	0
Global warming potential	675
Toxicity level	A (No toxicity identified)
Flammability level	Class 2 - Lower Flammability
Boiling Point:	-52.5°C
Critical Temperature	78.11°C
Critical Pressure	5.72 MPa
Auto ignition Temperature	648 °C
Flash Point	52°C (124°F)
Molar Mass	52.024 g·mol−1
Melting Point	-136 °C
Vapor pressure at 25°C	853.9 kPa
Density at 25°C	516.1 kg/m3

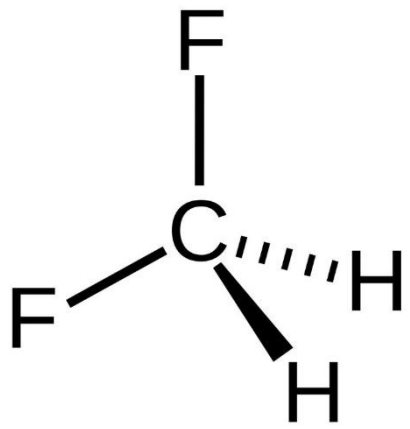

Figure 02: Structure of R32

2.1.2 Advantages and Disadvantages of R32

Smaller Impact on Environment

R-32 has zero ODP (Ozone Depletion Potential) and its GWP (Global Warming Potential) is 675, which is lower than the GWP of currently used R-410A or R-22. It could reduce the charging volume by 30% compared to R-410A. R-32 related CO2 emissions decrease by 76% thanks to the lower GWP and the charging volume reduction. While it has a lower GWP compared to older refrigerants, its GWP of 675 is still significant, prompting ongoing research into even greener alternatives.

Safety: Flammability

International Standard ISO 817:2014 segregates the flammability of refrigerants into 4 categories as follows: Class 1 (No flame propagation), Class 2L (Lower flammability), Class 2 (Flammable) and Class 3 (Higher flammability). Class 2L refrigerants present the lowest risk of the 3 flammable categories and are defined by having a burning velocity of less than 10 cm per second. The characteristic of this low burning velocity is that the flame front does not propagate readily in a horizontal direction. This is because the convection rise due to combustion creates a higher velocity than the burning velocity. This effectively means that a Class 2L refrigerant is not explosive if ignited because the flame only propagates in an upwards direction from the ignition point and not rapidly outwards in all directions.

Reduced Refrigerant Charge

Systems using R32 require less refrigerant by weight compared to alternatives, leading to cost savings and lighter designs.

Widely Adopted and Supported

It has become a standard refrigerant in many regions, ensuring availability, regulatory compliance, and manufacturer support.

Compatible with Existing Infrastructure

Many components and systems initially designed for R410A can be adapted for R32 with minimal changes.

Higher Operating Pressure

Works at higher pressures compared to some other refrigerants, necessitating robust and pressure-tested system designs.

Regulatory Restrictions in Some Regions

Some countries impose stricter regulations or require additional certifications due to its mild flammability.

2.2 R290 (Propane) as a Refrigerant

R290 (propane) is a natural hydrocarbon refrigerant increasingly used in air conditioning, refrigeration, and heat pump systems. It is known for its eco-friendly properties and excellent thermodynamic performance.

Applications of R290:

1. Domestic Refrigeration: Widely used in household refrigerators and freezers.
2. Air Conditioning Systems: Ideal for residential and light commercial air conditioning units.
3. Commercial Refrigeration: Common in supermarkets, beverage coolers, and vending machines.
4. Heat Pumps: Increasingly adopted in environmentally friendly heating and cooling solutions.

R290 is gaining traction as a sustainable refrigerant due to its low environmental impact and high efficiency. However, addressing flammability risks through better system designs and stricter safety standards is essential for its widespread adoption.

2.2.1 Features of R290

R290, also known as propane, is a colorless and odorless gas. It is a hydrocarbon with the chemical formula C3H8 and is classified as a flammable gas. R290 is used as a refrigerant in commercial and industrial refrigeration systems and as an aerosol propellant. R290 has a global warming potential (GWP) of 3. R290 is a hydrocarbon refrigerant.

R290 is one of several possible substitutes for the commonly used refrigerant R22. When used as a refrigerant, R290 has advantages over other options, such as R134a and R410a. For example, R290 is less expensive than R134a, and its GWP is much lower than R134a or R410a. However, due to its flammability, care must be taken when handling and using R290.

- Extremely Low Global Warming Potential (GWP): R-290 has an exceptionally low GWP of 3 over a 100-year period, making it one of the most environmentally friendly refrigerants available.
- Single Component: Similar to R-32, R-290 is a single-component refrigerant composed entirely of carbon and hydrogen atoms.
- Energy Efficiency: It offers high energy efficiency, contributing to reduced energy consumption and operational costs in refrigeration and air conditioning systems.
- Highly Flammable: Classified as highly flammable (A3) according to ASHRAE standards, R-290 requires strict safety measures due to its high flammability.
- Application in Domestic and Small Commercial Refrigeration: R-290 is commonly used in domestic refrigeration, vending machines, and small commercial refrigeration units due to its environmental benefits and efficiency.
- Limited Adoption in Larger Systems: Its high flammability restricts its adoption in larger air conditioning systems, where safety concerns are more significant.

Table 02: Features of R290

Classification	Hydrocarbon refrigerant
Chemistry	C3H8 (CH3CH2CH3)
Ozone depletion potential	0
Global warming potential	3
Toxicity level	A (No toxicity identified)
Flammability level	Class 3 – Highly flammable
Boiling Point:	230.90 to 231.11 K
Critical Temperature	96.7 °C (206.06 °F)
Critical Pressure	4,248 kPa
Auto ignition Temperature	470 °C (878 °F/ 743 K)

Flash Point	−104 °C (−155 °F/ 169 K)
Molar Mass	44.097 g·mol−1
Melting Point	−187.7 °C (−305.8 °F/ 85.5 K)
Vapor pressure at 25°C	551.2 kPa
Density at 25°C	958.2 kg/m3

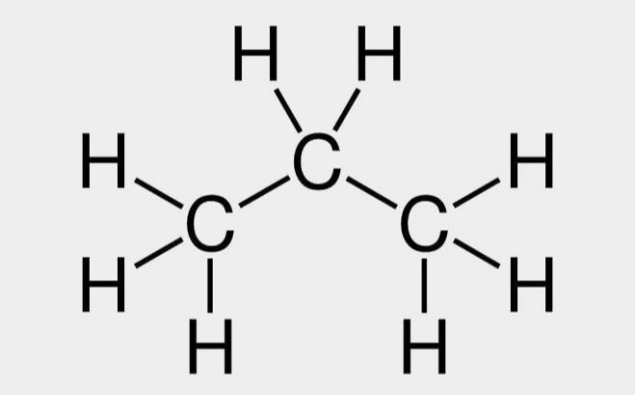

Figure 03: Structure of R290

2.2.2 Advantages and Disadvantages of R290

Advantages

- The largest attraction when it comes to using R-290 is its effect on the climate. Synthetic refrigerants such as CFCs, HCFCs, and HFCs all damage the environment. Some damage through Ozone depletion and others through Global Warming. Either way, they are harmful. Propane has zero Ozone depletion potential and has a Global Warming Potential of just three. In contrast, one of the most popular HFC refrigerants today, R-404A, has a GWP of nearly four-thousand. These facts alone are why the world is pushing for more and more R-290 applications.
- R-290 has excellent thermodynamic performance, it is energy efficient, and it is very reliable.
- Propane is very affordable and has ample supply especially when compared to some of the more expensive refrigerants out there like R-22.
- R290 is a naturally occurring refrigerant, making it environmentally friendly.

- It's less expensive than some other refrigerants on the market.
- In addition, it is a very energy efficient gas, making it a viable and sustainable alternative to synthetic refrigerant gases
- Contains no chlorine or fluorine, making it less harmful to the ozone layer.
- Compatibility with existing equipment.

Disadvantages

- The biggest drawback with propane, and with many other hydrocarbons, is flammability. Yes, I know most of you could have guessed that already since we're dealing with propane. The substance can be quite flammable when put under the right conditions. This is why it is rated as an A3 refrigerant from ASHRAE. The A standing for non-toxic and the 3 standing for 'higher flammability.'
- Because of this higher flammability risk with propane the amount of charges allowed by governments is quite limited. As an example, in the United States propane based systems cannot have a charge greater than one-hundred and fifty grams. This was actually recently changed by the EPA. (UL standard 60335-2-24 – Source) Before that the old limit was just fifty-seven grams. This rule change applied to refrigerators and freezers as well as other approved applications we'll get into further on into this article.
- Again, due to its flammability, R-290 is not suitable for use in retrofitting existing fluorocarbon based systems such as R-22, R-410A, or R-404A. These machines were not made to handle flammable refrigerants such as R-290.
- It has high pressure, so refrigeration systems using R290 must be designed accordingly to manage this pressure safely. This can add to the overall cost of the system.
- Requires additional safety measures such as ventilation and fire suppression systems.
- May have limited market availability compared to other refrigerant gases.
- May require modifications to existing equipment for use.

2.3 Basic Refrigeration Cycle

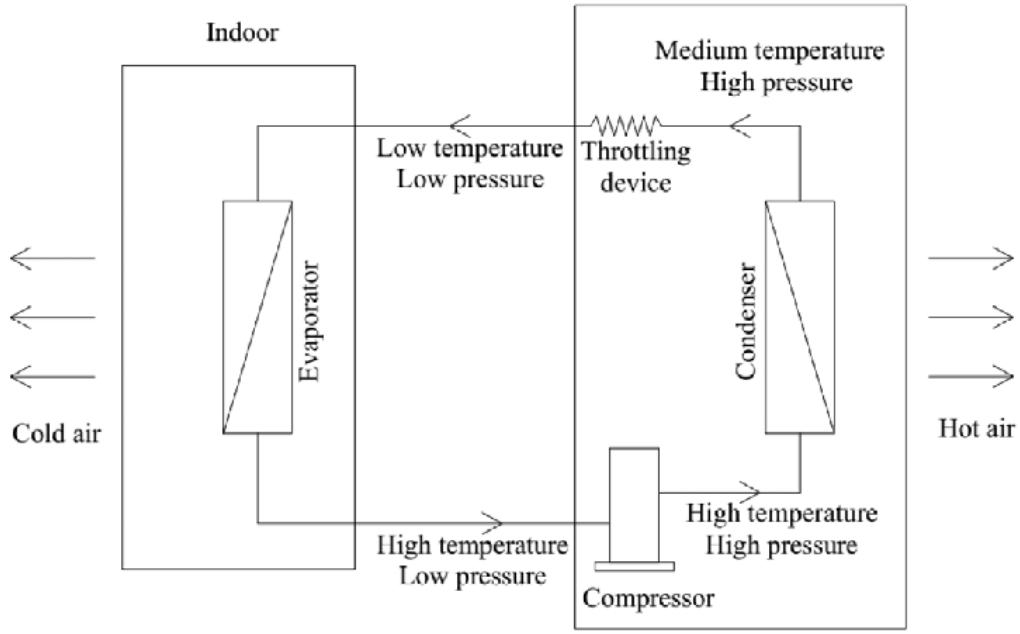

Figure 04: Refrigeration cycle in a split-type air conditioner

"Split system air conditioner" is a popular and effective means of cooling interior spaces. Its two main components, an outside unit and an indoor unit, are connected by refrigerant lines. Regardless of the weather, these devices work together to regulate the temperature and create a comfortable environment.

Inside the Split AC Unit

- The Indoor Unit: This is the part of the split AC you see inside the room. It is usually mounted high on a wall and contains the evaporator coil, a fan, and the air filter. When the desired temperature is set, the indoor unit starts working.
- The Outdoor Unit: This is installed outside the building. It consists of the compressor, condenser coil, and another fan. It is very significant in the process of exchanging heat.

The Cooling Cycle Begins

The whole process makes AC units very efficient in cooling down the indoor spaces. Whenever you switch on your split AC, the following processes occur:

- Extraction of Cool Air: The indoor unit fan starts circulating the warm air from the room over the evaporator coil. This coil has cold refrigerant within.
- Heat Absorption: While the warm air blows over the cold evaporator coil, the inside refrigerant absorbs all of the air's heat to turn it into a gas. The phase change is basically crucial to cool the air effectively.
- Cool Air Discharge: The by-now-cooled air is pushed back into the room through the indoor unit's fan. You begin to feel the relief of the chilled air, and the temperature of the room starts going down.2.4 Environmental Impact of R32

2.4 Environmental impact of R32

The environmental impact of R32 has both positive and negative sides, making it an intermediate solution in the transition toward sustainable refrigerants. On the positive side, R32 has zero ODP, meaning it does not harm the ozone layer. Moreover, its GWP is 675, which is way lower than the GWP of older refrigerants like R410A, which stands at approximately 2088. This reduction contributes to lessening climate change by reducing the carbon footprint from HVAC systems. In addition, R32 systems usually have higher energy efficiency, which indirectly protects the environment by reducing emissions related to the generation of electricity.

Despite these advantages, R32 still has environmental drawbacks. Its GWP of 675, while lower than many alternatives, is not negligible and contributes to global warming if leaked into the atmosphere. With an atmospheric lifetime of about 5 years, R32 persists in the environment before breaking down, which, although shorter than many other HFCs, still poses a concern. The refrigerant's reliance on synthetic production also places it behind natural refrigerants, such as R290 (propane) and R600a (isobutane), in terms of sustainability.

Compared to the alternatives, R32 is an improvement over R410A and R22 with considerably reduced environmental impact. However, natural refrigerants such as R290, boasting a GWP of just 3, and carbon dioxide (CO_2) with a GWP of 1, are much eco-friendlier. These options are increasingly prioritized in regions with stringent climate regulations, as they have no significant atmospheric lifetime or ozone-depleting effects.

R32 is the workhorse that has so far helped industries move towards low-GWP refrigerants in order to meet phasedown schedules under the Kigali Amendment to the Montreal Protocol. But as regulations continue to tighten and technologies improve, R32 has been considered a bridge toward ultra-low-GWP refrigerants, such as natural refrigerants or HFO-based solutions. While R32 reduces environmental impact compared to many of its predecessors, it is not the ultimate answer to the environmental challenges posed by refrigerants.

2.5 Environmental Impact of R290

R290, or propane, is considered one of the most environmentally friendly refrigerants because of its very low environmental impact. It has a GWP of only 3 and thus an almost negligible contribution to climate change compared with synthetic refrigerants such as R32, with a GWP of 675. Moreover, it has zero ODP, meaning that it does not damage the ozone layer. R290 is a non-toxic natural hydrocarbon refrigerant, and its ecological footprint is very small, making it a sustainable choice for refrigeration and air conditioning applications.

One of the most significant advantages of R290 is its short lifetime in the atmosphere since it breaks down very quickly when released into the air. This feature makes it harmless to long-term atmospheric warming, unlike R32, which stays in the environment for about five years. In addition, R290 systems typically require smaller refrigerant charges, reducing the environmental risk in case of leaks, making them safer for the environment.

Compared to R32, R290 has several advantages. Most importantly, it has a much lower GWP of 3; even in the case of leakage of the refrigerant, R290 has very negligible impacts on global warming. In contrast, R32, with its GWP of 675, has a much higher potential to contribute to climate change. Besides that, R290 is a naturally occurring hydrocarbon, hence making the processes of its production and waste management more ecological compared to synthetic refrigerants like R32. Both refrigerants have zero ODP; the natural composition of R290 coincides better with international ideas of sustainability and reduces the dependence on energy-intensive chemical processes while producing the product.

Another important benefit of R290 is that it also adheres to ever-evolving environmental regulations. Because international policies are trending toward ultra-low

GWP refrigerants, R290 will not need a phasedown or replacement in those countries that are using it. Its good thermodynamic properties also make for high energy efficiency comparable to or even better than R32 in systems that have been specifically optimized for hydrocarbon refrigerants.

R290 is much greener than R32, with ultralow GWP, a very short lifetime in the atmosphere, and a natural origin. It represents the ideal refrigerant choice in order to ensure sustainable cooling.

CHAPTER 3

PROJECT IDEA

3.1 Exciting Related Researches and Findings

1. Citation:

"Rizka Ulum, Berkah Fajar, Tony Suryo Utomo, "Experimental Study of Performance AC Split R32 Retrofitted with R290 at 50% Mass" Published in International Research Journal of Innovations in Engineering and Technology - IRJIET, Volume 6, Issue 7, pp 8-11, July 2022. Article DOI https://doi.org/10.47001/IRJIET/2022.607002".

Title: Experimental Study of Performance AC Split R32 Retrofitted with R290 at 50% Mass

Authors: Rizka Ulum, Berkah Fajar, Tony Suryo Utomo

Published in: International Research Journal of Innovations in Engineering and Technology (IRJIET), Volume 6, Issue 7, July 2022

This paper presents the investigation of R290 as a substitute refrigerant for R32 in a split-type air conditioner. R32 is one of the general refrigerants used in cooling systems, and it has been known to have a high GWP, which severely contributes to environmental concerns. On the other hand, R290 is an environment-friendly hydrocarbon refrigerant with a GWP of less than 3 and with no ODP. The research is intended to study the performance of R290 retrofitted at 50% mass of R32 in terms of cooling capacity, power consumption, and coefficient of performance.

The experiment was done using the split-type air conditioner of a 9000 BTU cooling capacity. Measuring the performance of the system was done at changing intensities of the room and environmental temperatures between 23-35°C. Parameters include cooling capacity, power consumption, and COP to identify the best comparison between both refrigerants.

The results showed that, due to the reduction of compression work, the cooling capacity of R290 was 34.33% lower than that of R32. However, the compressor power consumption of R290 is 40.24% lower than that of R32, which means large energy

saving. On the other hand, the COP of R290 is found to be 29.79% lower than R32, representing a drop in overall performance.

Therefore, while retrofitting R32 with R290 at 50% mass reduces the power consumption substantially and aligns well with environmental sustainability goals, there is a compromise on the issues of cooling capacity and efficiency. R290's performance has not been optimized for split-type air conditioners yet, and further research is needed for its improved application in practical systems. This again highlights the importance of a proper balance between environmental benefits and operational performance during the course of adopting alternative refrigerants.

2. Citation:

"IIR (2014). Performance of R32 and R290 as alternatives to R22 and R410A in domestic air conditioning. *International Institute of Refrigeration*. Available at: https://iifiir.org/en/news/performance-of-r32-and-r290-as-alternatives-to-r22-and-r410a-in-domestic-air-conditioning".

Title: Performance of R32 and R290 as Alternatives to R22 and R410A

The study has explored R32 and R290 as promising alternatives to conventional refrigerants R22 and R410A in air-conditioning systems. These refrigerants are evaluated in terms of performance and environmental benefits as a means of solving the pressing demand for sustainable cooling solutions with low GWP.

R32 had a higher cooling capacity than R290, R22, and R410A and, therefore, can be considered a good alternative to conventional refrigerants. R290 showed a lower cooling capacity compared to R22; however, it was found to perform well within the system's limit of operation, hence suitable for applications where such a reduction in capacity can be accommodated.

From the point of view of energy efficiency, R32 exhibited a higher COP than that of R22 and R410A, which means a better energy efficiency. R290 showed slightly lower COP values compared to R22 but still maintained good energy efficiency under specific conditions, making it competitive for energy-conscious applications.

From an environmental point of view, the performances of R32 and R290 were far better compared to R22 and R410A. R32 shows a much lower GWP value, while R290,

with an ODP of 0 and a GWP of less than 3, is exceptional. These metrics show that these refrigerants have more potential to minimize harm to the environment and contribute toward sustainability goals.

But the crucial point is safety considerations. The high flammability of R290 imposes strict safety measures that really limit its wide use. R32 is less flammable but also requires significant care; it is more balanced regarding performance, environmental friendliness, and safety for broad applications.

In the end, both R32 and R290 are viable alternatives to conventional refrigerants. R32 is more suitable for immediate, large-scale applications, while R290 can be adopted in specialized settings where its flammability can be effectively managed. These findings emphasize the need for tailored approaches to integrating alternative refrigerants into cooling systems.

3.2 Replacing R32 with R290 in a Same AC Unit

R32-based split AC machines find wide applications in residential spaces, commercial offices, and light industrial uses, offering a balanced blend of performance, energy efficiency, and environmental responsibility. Whereas R32 is a significant improvement in comparison with the previous generation of refrigerants, its Global Warming Potential is a factor in climate change and thus continues to be pursued toward ultra-low GWP alternatives or natural refrigerants. However, R32 continues to be one of the practical and effective choices of refrigerant for modern cooling applications.

R290-based Split AC machines mark a step ahead in sustainable cooling solutions. With ultra-low GWP, zero ODP, and excellent energy efficiency, this can be considered highly eco-friendly. While challenges like flammability and regulatory considerations exist, advancements in safety technologies and growing environmental awareness are driving the adoption of R290 systems. These machines are, in fact, better for the planet and come with considerably reduced operation costs, turning them into an excellent choice for both environmentally conscious customers and businesses.

R32-based split AC machines are suitable for regions that emphasize ease of use, safety, and immediate availability. They are highly energy-efficient with a moderate environmental impact, making them very popular as a transitional refrigerant. In contrast, systems with R290 will be more appropriate for environmentally sensitive

markets where sustainability and long-term regulatory compliance are a priority. While R290 has better energy efficiency and a minimal climate impact, its flammability has raised several safety challenges that require advanced handling and system design.

The adoption of R290 in a machine designed for R32 refrigerant is not easy, as there are big challenges regarding their physical and chemical properties, safety, and regulatory issues. While theoretically possible, retrofitting a system designed for R32 to use R290 is by no means straightforward; it requires extensive adjustments in the system components, design, and safety protocols.

R32 and R290 have different thermodynamic properties, meaning that the system that is optimized for one refrigerant will not work as effectively with the other. R32 is a synthetic refrigerant with a medium GWP of 675, while R290 is a natural refrigerant with a very low GWP of only 3. While both refrigerants are highly energy-efficient, R290 has better heat transfer properties that enable it to achieve better cooling with less energy. However, for R290, this system design may need to be completely changed by modifying the compressor, expansion valves, and evaporator/condenser coils, because R290 works under a different pressure and thermodynamic state of conditions than that of R32.

3.2.1 Environmental Impact

Various environmental impacts have to be considered when replacing R32 with R290 in an air conditioning system.

- Global warming potential
- Ozone depletion potential
- Moreover, during the replacement of refrigerants, energy efficiency of the new system should be considered, as it causes lowering indirect environmental impacts.
- Proper disposal and recycling of the old refrigerant (R32) and components from the replaced systems.

3.2.2 Safety Considerations

In the replacement of R32 with R290 in an air conditioner, several safety issues necessarily need to be addressed due to the flammability and special handling requirements of R290.

- Proper Ventilation: R290-based systems should be installed in well-ventilated areas so that in case of a leak, the refrigerant does not accumulate and cause any ignition.
- Leak Detection: Leak detection systems will help in quickly identifying the leakage of refrigerant, since even small quantities of R290 can create a fire hazard in congested areas.
- Safety Valves and Pressure Relief: R290 systems need safety valves and pressure relief devices that prevent over-pressurization with the purpose of reducing the risk of system failure or explosion.
- System Design Adaptations: The equipment-comprising compressors, piping, and electrical components-must be designed or specifically adapted to operate safely with R290, due to its higher flammability compared to R32.

Addressing such concerns will go a long way in making the transition to R290 safer by reducing its flammability-related risks and ensuring the product is at par with all safety standards.

3.2.3 Cost Comparison

When comparing the costs of replacing R32 with R290, the upfront costs, including equipment and installation, must be weighed against the operational costs, such as energy efficiency and maintenance. There are also long-term financial benefits that can be realized with the use of R290, including the significant reduction in GWP and potential regulatory incentives. On the other hand, safety requirements and special handling increase the cost of installation and servicing for R290. A balancing of these factors will provide a better insight into the overall cost-effectiveness of the replacement.

3.2.4 Thermodynamic Property Comparison

Comparing R32 and R290 based on thermodynamic properties, several factors have an effect on efficiency, performance, and suitability of the refrigerants for specific applications.

1. Cooling Capacity and Efficiency

2. Specific Heat Capacity

3. Saturation Pressure and Operating Pressure

4. Volumetric Cooling Capacity

5. Latent Heat of Vaporization

6. Energy Efficiency Ratio (EER)

7. Boiling Point and Operating Temperature Range

When replacing R32 with R290, the superior thermodynamic properties of R290, such as higher specific heat capacity, latent heat of vaporization, and thermal conductivity, generally make it a more efficient refrigerant in terms of energy consumption and cooling capacity. The lower operating pressures and better heat transfer characteristics of R290 contribute to its high efficiency and performance. However, the system must be carefully designed to handle R290's unique properties, particularly its flammability and lower pressure requirements, to ensure safe and optimal performance. R32, while still efficient, may not offer the same level of performance as R290, especially in terms of energy savings and cooling capacity.

3.2.5 Energy Efficiency

When comparing the energy efficiency of R32 and R290 as refrigerants in air conditioning systems, one usually refers to the EER or COP. The two parameters are used for evaluating how efficiently the refrigerant operates within a system.

1. Energy Efficiency Ratio (EER)

The Energy Efficiency Ratio is the ratio of cooling output in "BTU/h" or "kW" to a given unit of energy input in watts or kW. Hence, the higher the value of EER, the more effective the refrigerant/system will be at converting energy to cooling.

$$\mathrm{EER} = \frac{\text{Cooling Capacity (kW or BTU/h)}}{\text{Power input (kW or W)}}$$

2. Coefficient of Performance (COP)

The Coefficient of Performance (COP) is another key measure of energy efficiency, representing the ratio of useful heating or cooling energy output to the energy input. A higher COP indicates a more efficient system.

$$\text{COP} = \frac{\text{Cooling Output (W)}}{\text{Power input (W)}}$$

3. System Design and Energy Efficiency

While the inherent thermodynamic properties of refrigerants such as R32 and R290 are critical, system design is what determines overall energy efficiency.

Energy Consumption: In many cases, R290 systems consume less energy in real-world tests due to the better heat transfer of the refrigerant and lower operating pressures. These factors translate into reduced compressor work and more effective cooling cycles. As such, R290 systems typically exhibit a reduced overall energy consumption.

4. Energy Efficiency under Different Operating Conditions

The energy efficiency of R32 and R290 can vary depending on environmental conditions such as outdoor temperature, humidity, and internal system load.

R290 performs better in higher ambient temperatures due to its superior heat transfer properties, maintaining high cooling capacity and COP even in demanding conditions.

R32, while still efficient, may not perform as well in high ambient temperatures, which could result in slightly lower energy efficiency in such conditions.

- R290 basically outperforms R32 concerning energy efficiency since the thermodynamic properties of the former are better. These allow for lower energy use, therefore reducing operating costs. In any case, both refrigerants are quite efficient when compared to earlier ones, and the choice would rest on the respective systems designed and application requirements.

CHAPTER 4

RESULTS

4.1 Effect of Environmental Impact

When comparing R32 and R290 from an environmental impact point of view, several key factors need to be considered, including their Global Warming Potential (GWP), Ozone Depletion Potential (ODP), and overall environmental footprint throughout their life cycle, from production to disposal. A summary of the findings is presented in the following:

- Global Warming Potential (GWP):

R290, or propane, has a much lower GWP (about 3) than R32, whose GWP is 675. Because of this fact, R290 is much less harmful to the environment in case of release into the atmosphere because it results in a lesser contribution to global warming.

R32 is also more environmentally friendly than R22, but it still carries a much higher GWP compared to R290. Hence, in case of a leak, it causes global warming.

- Ozone Depletion Potential (ODP):

Both R32 and R290 have an ODP of 0, which means neither of them contributes to the depletion of the ozone layer. This makes both refrigerants safe from the point of view of ozone protection when used properly, unlike older refrigerants like R22, which had a higher ODP.

- Energy Efficiency and Indirect Environmental Impact:

R290 is more energy-efficient than R32, which implies that it takes less energy for R290-based air conditioners to achieve the same degree of cooling. Lower consumption of energy automatically means reducing carbon emissions during the life span of the system, especially from regions where electricity is produced from fossil fuels.

R32 is still efficient but typically consumes more energy than R290 for the same cooling capacity, which means it has a higher indirect environmental impact through increased energy usage and associated emissions.

- Regulatory Considerations:

With the Montreal Protocol and the Paris Agreement increasing their stringency on environmental laws worldwide, demand for low-GWP refrigerants such as R290 is expected to rise. The significantly lower GWP of R290 makes it more compliant with such regulations in the long term. R32 also complies with these regulations but will also suffer greater pressure as the world moves on to even lower GWP refrigerants.

- On the whole, from production, which is produced with fewer chemicals and energies, to disposal, with less hazard, R-290 is a better environmentally friendly refrigerant. Hence, it has lower overall environmental impact during its entire life in comparison to R32. This mostly applies to its very little GWP and its possible energy efficiency. Although R32 is more environmentally friendly than traditional refrigerants like R22, production and disposal of R32 are still more environmentally costly due to its higher GWP.

  R290 is, without a doubt, the greener refrigerant compared to R32, considering its lower GWP and energy efficiency. Though it has safety considerations due to its flammability, the environmental benefits of R290, especially in terms of reducing global warming potential and lowering energy consumption, make it an attractive option for the future of refrigerants.

  R290 is favorable considering the environmental impact.

4.2 Comparison of Safety Considerations

When comparing R32 and R290 in terms of safety considerations, both refrigerants present unique challenges and risks that need to be addressed.

- Flammability Risks

R290 (Propane) is classified as a highly flammable refrigerant (A3) under ASHRAE standards. This presents significant safety concerns related to the risk of fire or explosion during storage, handling, installation, operation, and maintenance. Special

precautions, such as proper ventilation, flameproof equipment, and adherence to strict safety standards, are required to mitigate these risks.

R32 is classified as mildly flammable (A2L), posing a much lower risk of ignition compared to R290. This makes R32 safer to use in environments where flammability concerns are critical. However, it is not completely non-flammable, so care is still necessary during use.

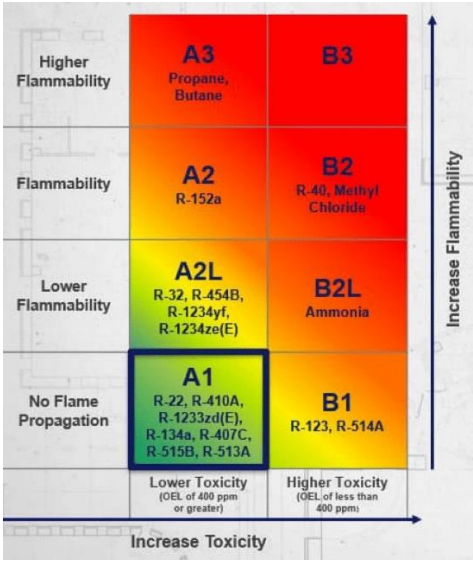

Figure 05: Classification of refrigerants according to flammability and toxicity

– Toxicity

Both R32 and R290 are considered low toxicity refrigerants (classified as Group A by ASHRAE), meaning they pose minimal risk to human health in normal concentrations. However,

* For R290, leaks in confined spaces could lead to the accumulation of gas, increasing the risk of fire and asphyxiation.
* For R32, leaks can lead to exposure to decomposition byproducts such as hydrogen fluoride in the event of combustion, which is toxic.

– Pressure and System Design

R32 operates at higher pressures compared to R290, which demands more robust system components to withstand the stress. Higher pressure increases the risk of system

failure if not properly maintained or designed. However, modern systems are built to handle these pressures safely.

R290 operates at lower pressures, which reduces mechanical stress on components and can lead to longer system lifespan. This makes R290 inherently safer in terms of pressure-related risks.

- Handling and Maintenance

R290 requires specialized training and handling procedures due to its flammability. Maintenance personnel must be certified and follow strict safety protocols, including the use of flameproof tools and leak detection systems.

R32, while easier to handle due to its lower flammability, still requires care to avoid ignition sources during leaks and proper disposal to prevent harmful environmental effects.

- Regulatory and Safety Standards

R290 systems must comply with strict safety standards due to their flammability, including charge limits (e.g., 150–500 grams in most applications), proper ventilation, and fire hazard mitigation. R32, being mildly flammable, has fewer restrictions and broader acceptance in various applications but still requires adherence to safety guidelines for A2L refrigerants.

- R32 is generally safer than R290 in terms of flammability, making it more suitable for applications where fire risk must be minimized. Its moderate operating pressure and mild flammability make it easier to handle and maintain compared to R290.

  R290, while highly flammable, offers safety advantages in terms of lower operating pressure and reduced mechanical stress on systems. However, its use demands strict adherence to safety protocols, training, and system design to mitigate risks.

  Ultimately, the choice between R32 and R290 depends on the specific application, environmental conditions, and the ability to manage safety concerns effectively.

While R32 may be preferred in applications prioritizing safety and ease of use, R290's environmental benefits and efficiency make it a strong candidate when safety protocols can be implemented.

4.3 Comparison of Cost

- Refrigerant Costs

R290 is significantly cheaper than R32 as a refrigerant due to its natural availability and simpler production process. This can lower the cost of refilling or replacing refrigerant after the initial adoption. R32, being a synthetic refrigerant, is more expensive per unit compared to R290, making the refrigerant cost higher for long-term use.

- Adopting R290 in a machine designed for R32 requires an initial higher investment for retro-fitting and safety improvements. However, the savings from lower refrigerant cost, reduced energy consumption, and compliance with future environmental regulations make it cost-effective over time.
  Continuing to use R32 will likely be more practical without much change in order to save costs immediately.

  However, R290 would have higher value in the long term, considering its better positions in terms of sustainability and energy savings.

4.4 Thermodynamic Properties

To proceed the replacement, we need to investigate the thermodynamic properties of both refrigerants and find out whether they are compatible with each other. Then only the same unit can work with both refrigerants.

1. The pressure variation against the temperature.

The figure 05 shows the variation of pressure against the temperature for both R290 and R32 refrigerants. With the increasing of temperature, the pressure of the refrigerants increases and the difference between two gases does not show a drastic change.

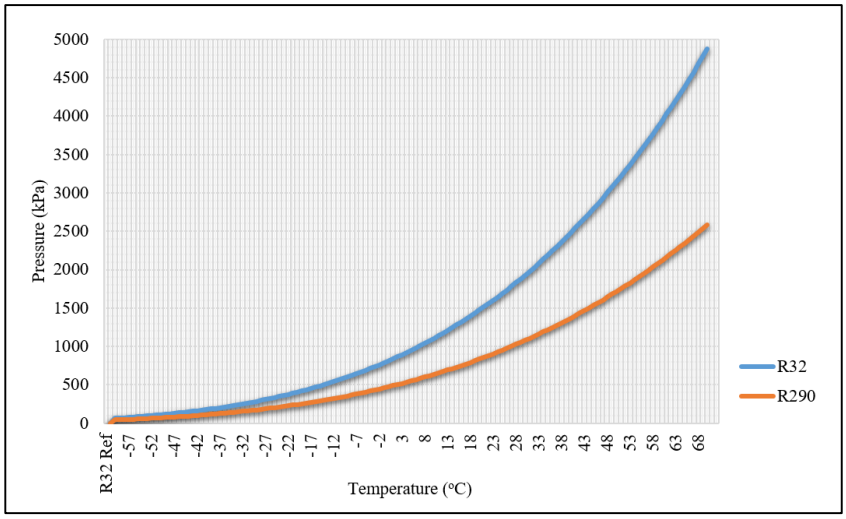

Figure 06: Variation of pressure (kPa) against the temperature (°C) for R32 and R290

2. Volumetric refrigerating effect

The VRE refers to the amount of cooling effect produced by one volume unit of the refrigerant circulated through the system. It is an important parameter used in the assessment of the efficiency of a refrigerant in systems like air conditioners and refrigeration units.

In other words, VRE indicates the amount of cooling capacity a refrigerant provides as compared to the volume of refrigerant handled by the compressor.

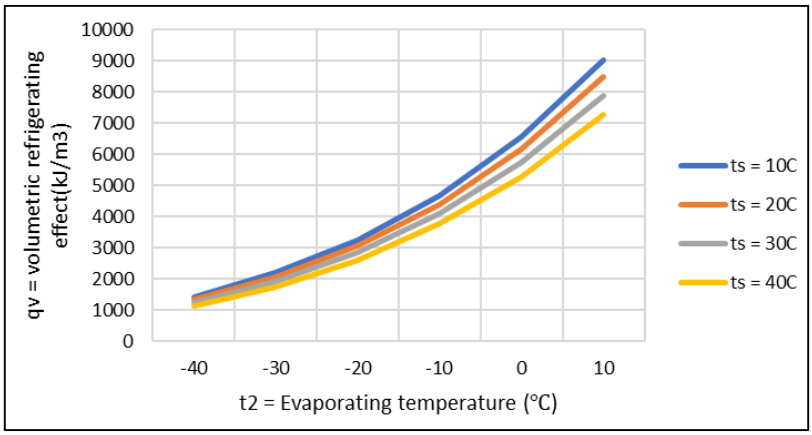

Figure 07: volumetric refrigerating effect against the evaporating temperature for R32

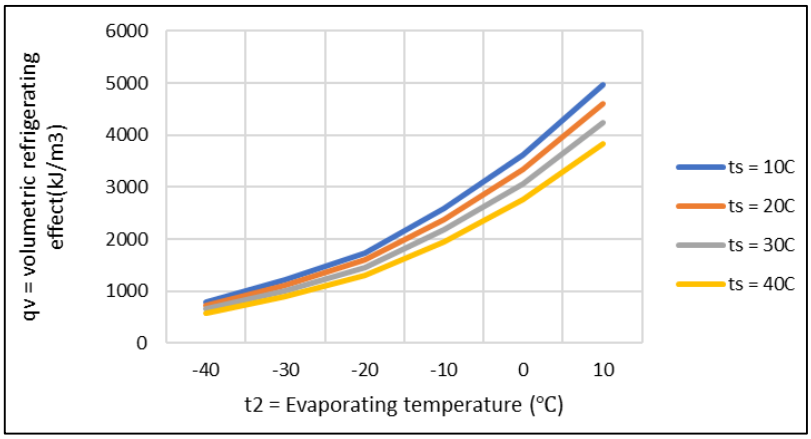

Figure 08: volumetric refrigerating effect against the evaporating temperature for R290

R32:

Higher Volumetric Refrigerating Effect (VRE):

R32 has a higher VRE than R290 across the range of evaporating temperatures due to its higher refrigerating capacity per unit of vapor volume. This makes R32 suitable for systems requiring compact compressors.

Performance Range: R32 typically performs better in systems designed for higher cooling capacities and works well in high-pressure systems.

Compressor Size: Smaller compressor size is sufficient for R32 compared to R290 for the same cooling capacity.

Better suited for compact systems requiring higher cooling capacity (e.g., residential AC systems).

R290:

Lower Volumetric Refrigerating Effect (VRE): R290 has a relatively lower VRE compared to R32 at the same evaporating temperatures. However, it still performs efficiently in terms of energy consumption (COP).

Eco-friendliness: Despite the lower VRE, R290 is favored for its low Global Warming Potential (GWP) and zero Ozone Depletion Potential (ODP).

Stability Across Range: R290's performance is consistent across a broader evaporating temperature range, making it ideal for applications where sustainability and lower environmental impact are prioritized.

Ideal for low-capacity cooling systems, particularly in eco-friendly applications (e.g., small commercial freezers, air conditioning in green buildings).

3. Variation of pressure ratio against the evaporating temperature.

The pressure ratio is an important parameter in thermodynamics and is typically calculated when analyzing compressors, turbines, or refrigeration systems. It is the ratio of the absolute discharge pressure to the absolute suction pressure of a system.

$$Pressure\ ratio\ = \frac{pressure_{discharge}}{pressure_{suction}}$$

Use pressure gauges installed on the compressor and measure discharge and suction pressures.

Normal Pressure Ratios: For air conditioning systems, typical pressure ratios range between 3 and 5, depending on the refrigerant and the operating conditions.

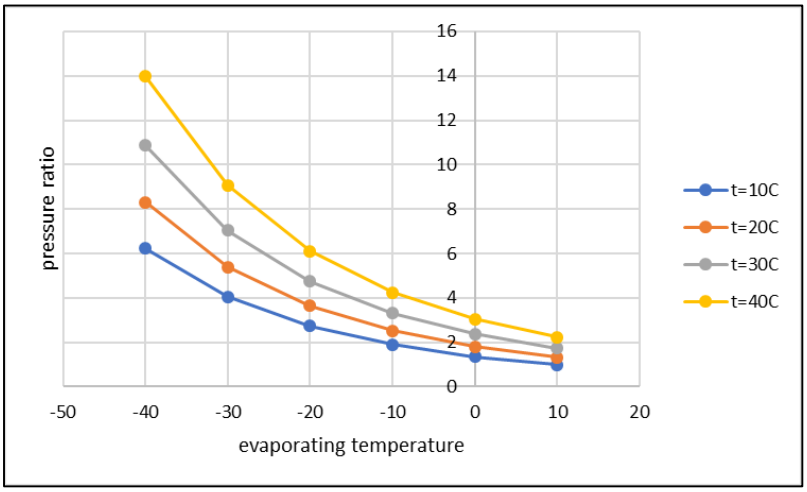

Figure 09: Variation of pressure ratio against the evaporating temperature for R32

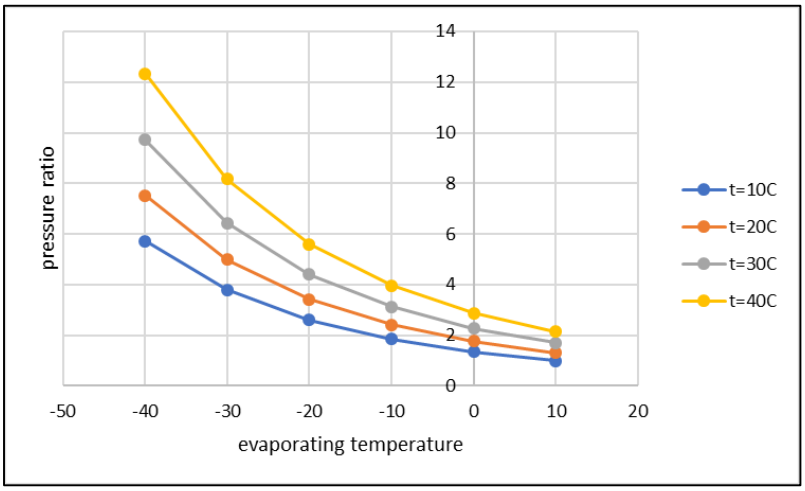

Figure 10: Variation of pressure ratio against the evaporating temperature for R290

R32: A high pressure ratio may indicate a heavily loaded system, poor efficiency, or high compression work.

R290: Lower pressure ratios indicate better system efficiency and lower compressor power consumption. Helps in identifying refrigerants that minimize compressor wear and tear.

4.5 Energy Efficiency

Energy efficiency is considered to be one of the major concerns when replacing R32 with R290 as a refrigerant in air conditioning systems.

– Thermodynamic Properties and Cooling Performance:

R290 has higher values for latent heat of vaporization and lower molecular weights when compared to R32, which means R290 absorbs more heat in case of phase change for unit mass. This fact directly corresponds to high energy efficiency with cooling performance.

– Volumetric Refrigeration Effect (VRE):

R32 has a higher VRE than R290, indicating that it offers good cooling capacity relative to its volume. However, many system designs compensate for this lower VRE with inherently superior energy efficiency in the case of R290. R290 systems generally

demand optimized compressor designs in order to realize its thermodynamic advantages.

- Energy Consumption by Compressor:

R290 has lower running pressure than R32, reducing the mechanical stress on the compressor and energy use. This means that energy costs during operation are low throughout the system's lifetime.

R32, while working at high pressures, can still achieve energy efficiency; it may require more power in compression, hence a slight increase in energy consumption.

- Heat Transfer Properties:

R290 exhibits superior heat transfer coefficients to R32, therefore enabling higher heat exchange effectiveness related to the evaporator and condenser. It improves systems' performances and energy consumptions.

R32 though effective does not achieve such grades of efficiency in heat transfers, which in turn forces the systems using this gas into higher energy consumptions.

- Environmental Benefits from Energy Efficiency:

The high energy efficiency in R290 means less indirect emission of greenhouse gases, especially when electricity is generated from fossil fuels.

Though R32 is energy efficient, it has a greater carbon footprint because of an increase in electricity use with a higher GWP when compared to R290.

- R290 enjoys better thermodynamic properties and lower operating pressures with superior heat transfer characteristics, hence more energy efficiency compared to R32.

  For the long term, these systems have lower consumption of electricity, which contributes towards reduced operation costs and minimizing environmental impacts. R32, on the other hand, is energy-efficient but slightly less efficient than R290, therefore leading to slightly higher energy consumption and resulting in emissions.

  In summary, replacing R32 with R290 offers significant energy efficiency gains, contributing to cost savings and environmental benefits.

CHAPTER 5

IDENTIFIED PRACTICAL PROBLEMS

Problem 1: Using the same compressor

The existing AC machine used in this project is a machine which works with R32 as its refrigerant. The refrigerant is filled into the compressor of the system. The compressor has a lubricant inside it to maintain its mechanical work. When the refrigerant is filled into the compressor, the lubricant gets mixed with the refrigerant, and subsequently, the viscosity of the refrigerant changes. Therefore, it is recommended that a specific compressor model should be filled with the refrigerant which is recommended by the designer.

The compressor model using in the existing machine is KSN108D32UFZ (R32). The lubricant inside is an ester type lubricant, which is a synthetic base oil that has been chemically synthesized.

Figure 11: Compressor of the AC machine

In order to refill the compressor with R290, R290 must be compatible with ester type lubricants. Referring the Table 02, we can identify that R290 can be used with ester type lubricants.

As describe above (Section 4.4) the pressure ratios of these two refrigerants does not vary drastically with the evaporating temperature, which means, R290 can be compressed using the same compressor as R32.

Therefore, we can refill the same compressor with R290 and run the AC machine.

Table 03: Lubricant types and viscosity ranges for different refrigerants

Refrigerant	Application	Type of Compressor	Lubricant Type	Lubricant ISO Viscosity
R-22, R-123, R-502	Home appliance, residential AC, auto AC, commercial and industrial AC and refrigeration	Reciprocating, rotary, scroll, centrifugal, screw	MO, POE, AB	32 to 320
R-134a	Home appliance, residential AC, auto AC, commercial and industrial AC and refrigeration	Reciprocating, rotary, scroll, centrifugal, screw	POE, PVE, PAG	7 to 220
R-407C	AC, heat pump	Scroll, reciprocating	POE	
R-404A/R-507	Stand-alone low-temperature coolers, ice machines, commercial and industrial refrigeration and freezers	Reciprocating, scroll, rotary	POE	32 to 68
R-410A	Residential and rooftop AC, heat pump	Scroll, reciprocating, rotary	POE, PVE, PAG	32 to 68
R-600/R-600a	Home appliance refrigeration Small stand-alone commercial refrigeration	Reciprocating, rotary, scroll	MO, POE, AB, PAO, PAG	7 to 68
R-290	Home appliance and small stand-alone commercial refrigeration	Reciprocating, scroll	MO, POE, AB	32 to 68
Ammonia (R-717)	Commercial and industrial refrigeration	Centrifugal, reciprocating, screw	PAG, AB, MO, PAO	32 to 220
Carbon dioxide (R-744)	Industrial and commercial refrigeration, heat pump hot-water heating, home appliance, beverage, mobile AC (car, bus, train)	Reciprocating, rotary, scroll	Subcritical: PAO, POE, PAG Transcritical: POE, PAG	68 to 120

*Examples presented are intended for general instruction and informational purposes only. Consult compressor and system maintenance and service manual for specific equipment for OEM approved lubricants.

MO – Mineral oil, POE – Polyol ester, AB – Alkybenzene, PAG – Polyalkylene glycol, PVE – Polyvinyl ether

Problem 2: Lower cooling capacity of R290

As explained in the section 4.4 the volumetric cooling capacity of R290 is slightly lower than R32. This causes the reduction of the efficiency of AC machine. When the machine is refilled with R290, it will consume more time to achieve the desired temperature than R32. Which means the rated BTU value of the machine cannot be achieved.

The experimental analysis is limited for domestic applications. Therefore, the cooling capacity of the AC machine used here is smaller and the time consumption would not have a significant variation compared to original R32 machine when considering the small domestic applications. Otherwise, when the replacement is done for large industrial applications, suitable changes in the compression work must be applied in order to maintain the required cooling capacity.

Problem 3: Higher flammability of R290

As explained in section 4.3, R290 is a highly flammable refrigerant with an auto ignition temperature of 470 °C (878 °F/ 743 K). Referring to figure 05, R290 works at lower pressures than R32 for same evaporating temperature, which means that R290 operates under lower pressures, reducing the risk of flammability.

Problem 4: COP value

Calculation of COP values for two cases assuming,

- Condenser temperature is 40 ^{0}C.
- Evaporator temperature is 10 ^{0}C.
- Process 1 – 2 is isentropic.

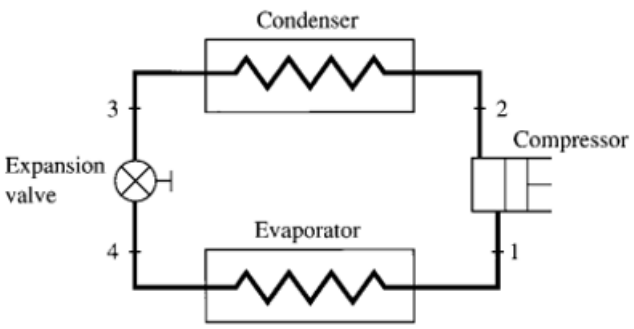

Figure 12: Schematic representation of a basic refrigeration system

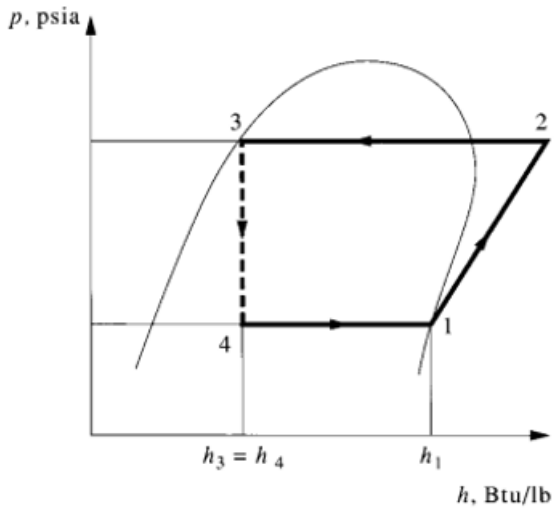

Figure 13: Pressure Vs. Enthalpy diagram of a refrigeration cycle

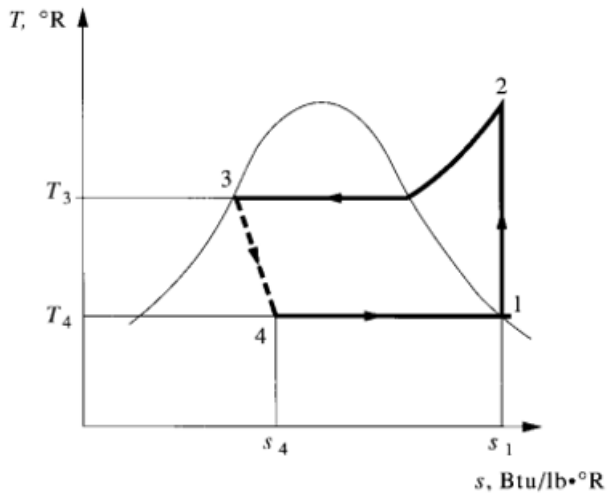

Figure 14: Temperature Vs. Entropy diagram for refrigeration cycle

COP value considering R32 as the refrigerant.

P_1 = 1.10690 MPa, P_3 = 2.47830 MPa, h_1 = 516.66 kJ/kg, h_2 =553 kJ/kg
h_3 = 275.61 kJ/kg, S_1 = 2.1185 kJ/kg.K, S_3 = 1.2520 kJ/kg.K

$P_1 = P_4$

$P_3 = P_2$

$S_1 = S_2$

$h_3 = h_4$

Work input = $h_2 - h_1$
= 553 KJ/Kg - 516.66 KJ/Kg
= 36.34 KJ/Kg

Q_{in} = $h_1 - h_4$
= 516.66 KJ/Kg- 275.61 KJ/Kg
= 241.01 KJ/Kg

COP = Q_{in} / Work input
= 6.633

COP value considering R290 as the refrigerant.

P_1 = 0.63660MPa, P_3 = 1.3694 MPa, h_1 = 585.67 kJ/kg, h_2 = 620 kJ/kg
h_3 = 307.15 kJ/kg, S_1 = 2.3626 kJ/kg.K, S_3 = 1.3594 kJ/kg.K

$P_1 = P_4$

$P_3 = P_2$

$S_1 = S_2$

$h_3 = h_4$

Work input $= h_2 - h_1$
$= 620$ KJ/Kg - 585.67KJ/Kg
$= 34.33$ KJ/Kg

Q_{in} $= h_1 - h_4$
$= 585.67$KJ/Kg- 1.3694 KJ/Kg
$= 278.52$ KJ/Kg

COP $= Q_{in}$ / Work input
$= 8.113026$

∴ COP R-32 = 6.633 < COP R-290 = 8.113026

Above calculation for COP value cannot be adopted practically as, the operating conditions of R290 are not as same as the operating conditions of R32. Therefore, we need to consider the power consumption for each case, i.e. the pressure ratio.

The lower the pressure ratio the lower the power consumption. Figure 08 and figure 09 show that the pressure ratios of R290 are lower than R32. Therefore, the power consumption of R290 is less than R32, resulting higher efficiency with R290 as the refrigerant.

CONCLUSION

The comparison of R32 and R290 as refrigerants for a domestic split air conditioning system shows huge differences in their environmental impact, safety issues, cost, and efficiency of energy consumption. At this point, R290 is an environmentally friendly alternative because it has a GWP of nearly zero and a zero ODP, which is opposed to the average GWP of R32. However, the flammability of R290 would require extra safety measures: flameproof components, ventilation, and leak detection systems.

From the point of view of energy efficiency, R290 outperforms R32 owing to its thermodynamic properties, such as higher latent heat of vaporization and better heat transfer coefficients. This means lower energy consumption and reduced operating costs over time. Although the initial costs of retrofitting systems for R290 may be a bit higher due to necessary safety modifications, its long-term energy savings and environmental benefits make it cost-effective.

As a conclusion, the replacement of R32 with R290 will be feasible and viable if all the safety measures are taken seriously. The shift agrees with the movement on using environmentally friendly refrigerants being advanced globally.

FUTURE WORK

The future scope of this project is to practically apply the replacement of R32 with R290 in a domestic split air conditioning system and to make detailed performance measurements. Implementation will be done in the thermodynamics lab by using a climate chamber to simulate controlled environmental conditions. During the running of the systems, key data such as condenser and evaporator pressures and temperatures will be recorded. It will calculate the cooling capacity in BTU/hr, depending upon the time taken by both refrigerants for a fixed temperature difference. The experiment will therefore provide direct comparisons of the energy efficiency and performance of R290 and the practical feasibility of its application as a substitute for R32 with some useful insight into its potential as an alternative refrigerant that is environmentally friendly and efficient.

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APPENDICES

Table: Volumetric refrigerating effect of different evaporating temperature of refrigerant R32

R 32				
ts=10 c				
t2	h2	hs	v2	qv
-40	502.02	217.74	0.19743	1439.90275
-30	506.27	217.74	0.13091	2204.03331
-20	509.97	217.74	0.08963	3260.40388
-10	513.02	217.74	0.06301	4686.24028
0	515.3	217.74	0.04527	6573.00641
10	516.66	217.74	0.03308	9036.2757
ts=20c				
-40	502.02	236.12	0.19743	1346.80646
-30	506.27	236.12	0.13091	2063.6315
-20	509.97	236.12	0.08963	3055.33861
-10	513.02	236.12	0.06301	4394.54055
0	515.3	236.12	0.04527	6166.99801
10	516.66	236.12	0.03308	8480.65296
ts=30c				
-40	502.02	255.32	0.19743	1249.5568
-30	506.27	255.32	0.13091	1916.96585
-20	509.97	255.32	0.08963	2841.12462
-10	513.02	255.32	0.06301	4089.82701
0	515.3	255.32	0.04527	5742.87608
10	516.66	255.32	0.03308	7900.24184
ts=40c				
-40	502.02	275.61	0.19743	1146.7862
-30	506.27	275.61	0.13091	1761.97388
-20	509.97	275.61	0.08963	2614.74953
-10	513.02	275.61	0.06301	3767.81463
0	515.3	275.61	0.04527	5294.67639
10	516.66	275.61	0.03308	7286.88029

ts= inlet temperature to the expansion device

Table: Volumetric refrigerating effect of different evaporating temperature of refrigerant R290

R 290				
ts=10c				
t2	h2	hs	v2	qv
-40	528.48	225.4	0.37985	797.8939
-30	540.38	225.4	0.25861	1217.973
-20	552.13	225.4	0.18767	1740.982
-10	563.65	225.4	0.13103	2581.47
0	574.87	225.4	0.09661	3617.327
10	585.67	225.4	0.07255	4965.817
ts=20c				
t2	h2	hs	v2	qv
-40	528.48	251.64	0.37985	728.814
-30	540.38	251.64	0.25861	1116.507
-20	552.13	251.64	0.18767	1601.162
-10	563.65	251.64	0.13103	2381.21
0	574.87	251.64	0.09661	3345.72
10	585.67	251.64	0.07255	4604.135
ts=30c				
t2	h2	hs	v2	qv
-40	528.48	278.83	0.37985	657.2331
-30	540.38	278.83	0.25861	1011.368
-20	552.13	278.83	0.18767	1456.28
-10	563.65	278.83	0.13103	2173.701
0	574.87	278.83	0.09661	3064.279
10	585.67	278.83	0.07255	4229.359
ts=40c				
-40	528.48	307.15	0.37985	582.6774
-30	540.38	307.15	0.25861	901.8599
-20	552.13	307.15	0.18767	1305.376
-10	563.65	307.15	0.13103	1957.567
0	574.87	307.15	0.09661	2771.142
10	585.67	307.15	0.07255	3839.008

Table: Pressure ration Vs. Evaporating temperature for R290

R290 condenser tem=10C			
evaporating temperature	evaporator pressure	condenser pressure	pressure ratio
-40	0.11112	0.6366	5.728941685
-30	0.16783	0.6366	3.793123995
-20	0.24452	0.6366	2.603468019
-10	0.34528	0.6366	1.843721038
0	0.47446	0.6366	1.341735868
10	0.6366	0.6366	1
R290 condenser tem=20C			
-40	0.11112	0.83646	7.527537797
-30	0.16783	0.83646	4.983971876
-20	0.24452	0.83646	3.420824472
-10	0.34528	0.83646	2.422555607
0	0.47446	0.83646	1.762972643
10	0.6366	0.83646	1.313949105
R290 condenser tem=30C			
-40	0.11112	1.079	9.710223182
-30	0.16783	1.079	6.42912471
-20	0.24452	1.079	4.412726975
-10	0.34528	1.079	3.125
0	0.47446	1.079	2.274164313
10	0.6366	1.079	1.694941879
R290 condenser tem=40C			
-40	0.11112	1.3694	12.32361411
-30	0.16783	1.3694	8.15944706
-20	0.24452	1.3694	5.600359889
-10	0.34528	1.3694	3.966056534
0	0.47446	1.3694	2.886228555
10	0.6366	1.3694	2.1511153

Table: Pressure ration Vs. Evaporating temperature for R32

R32 condenser tem=10C			
evaporating tem	evaporator pressure	condenser pressure	pressure ratio
-40	0.17741	1.1069	6.239219886
-30	0.27344	1.1069	4.048054418
-20	0.40575	1.1069	2.728034504
-10	0.58263	1.1069	1.899833514
0	0.8131	1.1069	1.361333169
10	1.1069	1.1069	1
R32 condenser tem=20C			
-40	0.17741	1.4746	8.311820078
-30	0.27344	1.4746	5.392773552
-20	0.40575	1.4746	3.634257548
-10	0.58263	1.4746	2.530937302
0	0.8131	1.4746	1.813553069
10	1.1069	1.4746	1.332188996
R32 condenser tem=30C			
-40	0.17741	1.9275	10.86466377
-30	0.27344	1.9275	7.049078408
-20	0.40575	1.9275	4.750462107
-10	0.58263	1.9275	3.308274548
0	0.8131	1.9275	2.370557127
10	1.1069	1.9275	1.741349715
R32 condenser tem=40C			
-40	0.17741	2.4783	13.96933657
-30	0.27344	2.4783	9.063414277
-20	0.40575	2.4783	6.107948244
-10	0.58263	2.4783	4.253642964
0	0.8131	2.4783	3.04796458
10	1.1069	2.4783	2.238955642

Table: Variation of pressure against temperature

Temp (°C)	Pressure (kPa)		Temp (°C)	Pressure (kPa)		Temp (°C)	Pressure (kPa)		Temp (°C)	Pressure (kPa)	
	R32	R290		R32	R290		R32	R290		R32	R290
-60	64.96	42.7	-28	296.7	181.4	5	951.4	551.1	38	2360	1307
-59	68.65	45	-27	309	188.6	6	981.1	567.5	39	2418	1338
-58	72.52	47.4	-26	321.6	195.9	7	1012	584.2	40	2478	1369
-57	76.56	49.9	-25	334.6	203.4	8	1043	601.3	41	2539	1401
-56	80.77	52.2	-24	348	211.2	9	1074	618.8	42	2601	1434
-55	85.17	55.2	-23	361.8	219.2	10	1107	636.6	43	2665	1467
-54	89.76	58.1	-22	376	227.4	11	1140	654.8	44	2729	1500
-53	94.55	61	-21	390.7	235.8	12	1174	673.4	45	2795	1534
-52	99.54	64.1	-20	405.8	244.5	13	1209	692.4	46	2862	1569
-51	104.7	67.3	-19	421.3	253.4	14	1244	711.8	47	2930	1604
-50	110.1	70.6	-18	437.3	262.6	15	1281	731.5	48	2999	1640
-49	115.8	74	-17	453.7	272	16	1318	751.7	49	3069	1676
-48	121.6	77.6	-16	470.7	281.7	17	1356	772.3	50	3141	1713
-47	127.7	81.3	-15	488.1	291.6	18	1395	793.2	51	3214	1751
-46	134	85.1	-14	506	301.8	19	1434	814.6	52	3289	1789
-45	140.6	89.1	-13	524.4	312.3	20	1475	836.5	53	3364	1828
-44	147.4	93.2	-12	543.3	323	21	1516	858.7	54	3441	1867
-43	154.5	97.4	-11	562.7	334	22	1558	881.4	55	3520	1907
-42	161.9	101.8	-10	582.6	345.3	23	1601	904.5	56	3600	1948
-41	169.5	106.4	-9	603.1	356.8	24	1645	928.1	57	3681	1989
-40	177.4	111.1	-8	624.1	368.7	25	1690	952.1	58	3764	2031
-39	185.6	116	-7	645.7	380.9	26	1735	976.5	59	3848	2074
-38	194.1	121.1	-6	667.9	393.3	27	1782	1001	60	3933	2117
-37	202.9	126.3	-5	690.6	406	28	1830	1027	61	4020	2161
-36	212	131.7	-4	713.9	419.1	29	1878	1053	62	4109	2205
-35	221.4	137.2	-3	737.8	432.5	30	1928	1079	63	4199	2250
-34	231.1	143	-2	762.3	446.1	31	1978	1106	64	4291	2296
-33	241.2	148.9	-1	787.4	460.1	32	2029	1133	65	4384	2343

-32	251.6	155	0	813.1	474.5	33	2082	1161	66	4479	2390
-31	262.3	161.3	1	839.5	489.1	34	2135	1189	67	4576	2438
-30	273.4	167.8	2	866.5	504.1	35	2190	1218	68	4675	2487
-29	284.9	174.5	3	894.1	519.4	36	2245	1247	69	4775	2537
			4	922.5	535.1	37	2302	1277	70	4877	2587

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Experimental Study of Performance AC Split R32 Retrofitted with R290 at 50% Mass

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***Abstract* - Split air conditioners are now widely used by the general public as air conditioners, but the refrigerants used, one of which is the refrigerant R32, are still not environmentally friendly. The AC machine used in this study uses R32 before replacing it with R290, which has a capacity of 1 PK and is installed in a duct. The purpose of this research is to evaluate the performance of the low-GWP refrigerant R290. The mass of R290 charged to the AC engine is 50%, and temperature variations resulted in a 40.24 percent reduction in power consumption from R290 to R32. R290 has a 34.33 percent lower cooling capacity than R32 due to the lower cooling capacity, which affects the compressor's lower power consumption. From R290 to R32, the COP decreased by 29.74%. Overall, the performance of replacing R32 refrigerant with R290 refrigerant has not been optimized.**

Keywords: Energy Conversion, Air Conditioning, R32, R290.

I. INTRODUCTION

Cooling machine technology is currently having a significant impact on modern life, one of which is split AC, which is an air conditioner that is commonly found in offices, homes, and commercial spaces such as restaurants, supermarkets, hotels, apartments, and hospitals[1]. This tool is extremely important for creating thermal comfort for people in a room and effectively supporting walking activities, not only for improving the quality and comfort of life but also for touching the essential things that support human life[2].

Global warming and the destruction of the ozone layer are two of the most pressing environmental issues facing the world today. The destruction or depletion of the ozone layer is caused by Ozone Depleting Substances (ODS), which are emitted from a variety of activities, including the use of or production of goods containing ODS. Chlorofluorocarbons (CFCs) and hydrochloro carbons are two of the known threats to the ozone layer (HFCs). CFC refrigerants deplete the ozone layer, and HFC refrigerants emit greenhouse gases. Chlorofluorocarbons (CFCs) are synthetic chemical compounds that are commonly used as refrigerants in cooling systems[3].

Chlorofluorocarbons (CFCs) are synthetic chemical compounds used in cooling systems, also known as refrigerants. There are many different types of refrigerants used in cooling systems, one of which is HCFC (Hydro-Chloro-Fluoro-Carbon) refrigerant, an example of which is R22 (HCFC-22). Because this ozone-depleting chemical is very stable, it can reach the stratosphere in its entirety. The ozone layer in the stratosphere serves to prevent ultraviolet-B rays from reaching the earth's surface. When these chemicals reach the stratosphere, they are converted by ultraviolet radiation from the sun into ozone-depleting chlorine atoms[4].

The use of environmentally friendly refrigerants is critical to supporting government programs aimed at reducing the impact of ODS on the ozone layer and materials with a high GWP. According to the 1987 Montreal Protocol, which was later revised by the London Amendment in 1990 and again by the Copenhagen Amendment in 1992, the use of CFC 12 and HCFC 22 in the world must be limited in order to be declared as prohibited refrigerants[5].

Alternative refrigerants include hydrocarbon refrigerants. Hydrocarbon refrigerant is a refrigerant that is currently being researched because it is environmentally friendly, non-toxic, cheaper, does not cause ozone depletion (ODP of 0), and does not cause global warming (GWP of less than 3) R290 is a natural material that has zero ODP and low to negligible GWP. It is not harmful to the environment[6].

Many studies have shown that the R290 performs on par with or even better than the R22. Furthermore, R290 is highly compatible with HVAC materials and lubricants, as well as being low-cost and widely available. The only issue with R290 is that, like any other flammable substance, its high flammability can lead to dangerous incidents if not handled properly. As a result, the international community has always taken it seriously and banned its use[7].

Berkah and colleagues'renovation a small-capacity air conditioner from R22 to R290. The results showed that the maximum improvement in AC performance was obtained when the mass of R290 was 50% of that of R22. Cooling capacity decreased by 2.8 percent while EER (energy

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efficiency ratio) increased to 12.6 percent at this optimal charge[8].

A study using the refrigerant R22 and a window air conditioner with a cooling capacity of 2.4 kW and an energy efficiency ratio (EER) of 3.2 installing a compressor with a capacity that is 20% greater to fill R290. The results show that the R290 will perform better if it adopts a larger displacement compressor at the same time. When the outdoor temperature drops, all alternative systems have higher upgrade rates and greater increases in cooling capacity and EER than the R22 system. The R290 system has significantly increased cooling capacity while decreasing EER[9].

The efficiency of a split-type air system using R22 and R290 The results show that the mass flow rate for R290 is only 44% and 47% when compared to R22, which is due to different mass density factors. When compared to R22, the cooling capacity of the system using R290 is 4.7 percent to 6.7 percent lower, as is the input power, but the EER of R290 is 8.5 percent higher, demonstrating one of the benefits of using R290[10].

An experimental retrofit study on a domestic split type AC system with R-32 inverter refrigerant using a 70 percent: 30 percent by weight mixture of R-32 and R290 refrigerant without system replacement. A comparison of the performance or COP of mixed refrigerants R32 and R290 reveals a 14 percent increase in COP and a 47 percent decrease in power usage when compared to the use of refrigerant R32, but a 33 percent decrease in cooling capacity[11].

II. EXPERIMENTAL DEVICE

2.1 Model Test

The experimental equipment is a split-type air conditioning unit with a cooling capacity of 9000 BTU. A split air conditioner consists of a compressor, condenser, evaporator, and capillary tube. After this major component is removed from its original design, the condenser and evaporator are installed in a split-AC unit. Because of the component's connection to the refrigerant pipe, performance data for the compressor, condenser, and evaporator are easily accessible. On each component, temperature and pressure are recorded.

By measuring the refrigerant flowing in the capillary tube, the mass flow rate of the refrigerant in the system is calculated. Compressor power is measured using electric current, voltage, and power factor. Data acquisition devices and measurement instruments include desktop computers, data acquisition (DAQ NI), temperature sensors (LM35), pressure gauges, and digital scales. Figure 1 depicts the universal testing machine and measuring instruments used in the experiment.

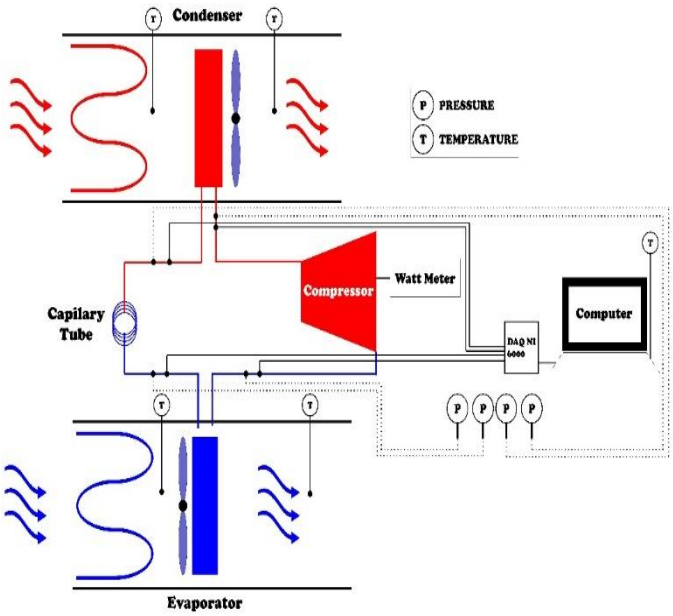

Figure 1: Schematic diagram

2.2 The Procedure for the Experiment

The experiment was conducted in two stages, the first using a split AC filled with R32 and the second with R290. The experiment began with an R32 mass of 430 grams (according to the label AC R32). Experiments were carried out at temperatures ranging from 23°C to 30°C, 25°C to 30°C, (27°C to 30°C), (25°C to 35°C), and (27°C to 35°C). R290 is filled with 50% (215 grams) of the total mass of R32, in accordance with the hydrocarbon filling reference (MC22/R290).

With a 5-minute delay between measurements, the measurements were taken after the system had been stable for 60 minutes. The experiment starts with 430 grams of mass. The vacuum process is repeated after the R32 refrigerant data collection system has been completed.

III. RESULTS AND DISCUSSIONS

3.1 Cooling Capacity

The cooling capacity of R32 is compared to that of R290 in Figure 2. The cooling capacity of an air conditioner is a measurement of its ability to dissipate heat. The greater the cooling capability, the sooner the room temperature rises to the required level. R290 has a lower cooling capacity than R32 due to a decrease in compression work (Figure 2).

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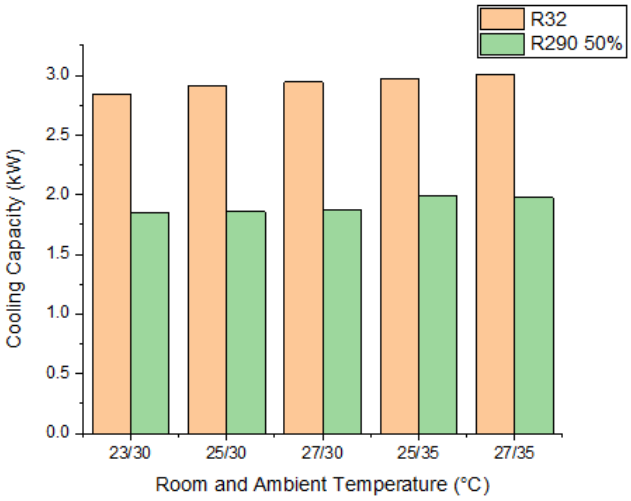

Figure 2: Cooling capacity

3.2 Power Consumption

Figure 3 shows that the power consumption of the R290 refrigerant compressor is lower than that of the R32 refrigerant compressor at all room and environmental temperatures. This is because R290 has a lower mass than R32. Because the compressor's work becomes lighter, less power is required.

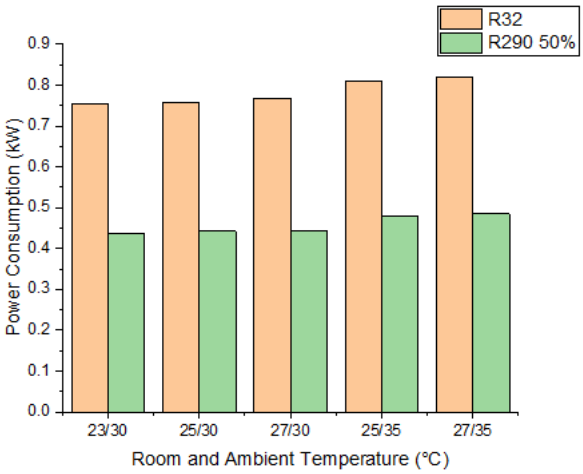

Figure 3: Power consumption

3.3 Coefficient of Performance

Figure 4 depicts the performance coefficients of the refrigerants R32 and R290. The diagram shows a drop in COP. Overall, the results show that the COP of refrigerant R290 is lower than the COP of R290 for all room and environmental temperature conditions. The cooling capability and power consumption of R32, which is greater than that of R290, have an impact on the COP value of each refrigerant used.

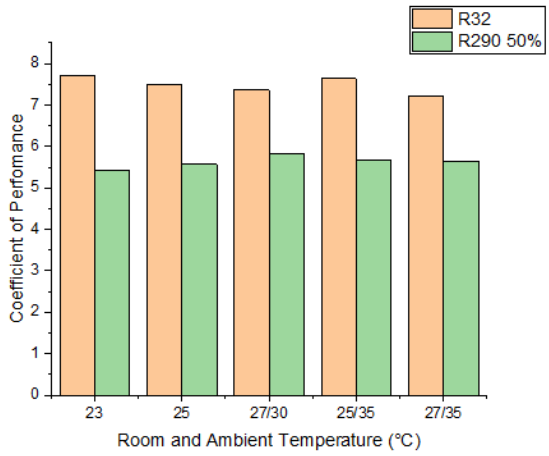

Figure 4: Coefficient of performance

IV. CONCLUSION

According to the findings of this study, the use of refrigerant R290 retrofitted R32 with a mass of 50% and variations in room temperature and ambient temperature resulted in a 40.24%decrease in power consumption from R290 to R32. The cooling capacity of R290 is 34.33% lower than that of R32 because the cooling capacity is lower, affecting the compressor's lower power consumption. The COP decreased by 29.79 % from R290 to R32. Overall, the performance of using R290 refrigerant to replace R32 refrigerant has not been optimized.

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Performance of R32 and R290 as alternatives to R22 and R410A in domestic air conditioning

According to the authors, regardless of the capacity difference, R290 could be considered as the most appropriate substitute in air conditioners with small finned tube heat exchangers

In order to identify alternative refrigerants exhibiting favorable cycle performance in domestic air conditioning, S. Cheng et al.* assessed the performance of R32 and R290 as substitutes to R22 and R410A in a small split household air conditioner.

The experiments were conducted with a small (5-mm diameter) finned tube heat exchanger – allowing a decrease in the quantity and cost of copper used – based on the enthalpy method. The results showed that in nominal cooling conditions, the COP (coefficient of performance) of R32 and R290 were 26.8% and 20.4% higher than R22, 7.3% and 2.1% higher than R410A. And in nominal heating conditions, the COP of R32 and R290 were both 11.0% higher than R22, 5.3% higher than R410A.

The systems with R290 and R32 had similar capacities to that with R22 and R410A in heating mode, but a huge difference of capacities in cooling mode: the capacity with R290 was similar to that with R22 and R410A but the capacity with R32 was 20.7% and 4.2% higher.

According to the authors, regardless of the capacity difference, R290 could be considered as the most appropriate substitute in air conditioners with small finned tube heat exchangers due to its lowest optimal charge quantity (335g) which is even less than the half amount of the other three, as low refrigerant charge represents reduction in the volume of the compressor, evaporator and condenser.

Aiming at further lowering the charge of flammable R290, a suction line heat exchanger was used; the cycle performance was enhanced: +5.3% in capacity and +5.1% in COP; safety was also improved with a 6% reduction of the optimal charge (315g).

Cycle performance of alternative refrigerants for domestic air-conditioning system based on a small finned tube heat exchanger, Song Chen et al., Applied Thermal Engineering 64 (2014) 83-92

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Themes : Other air-conditioning equipment;

HFCs

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Commissions : Air conditioning